

ROXANNE LYLE

DATA ANALYTICS & AI | PYTHON | MACHINE LEARNING | NLP

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PROFESSIONAL SUMMARY

B.S. Information Technology student specializing in Data Analytics and Artificial Intelligence with a 4.0 GPA and applied project experience in Python, machine learning, generative AI/NLP, explainable AI, responsible AI, forecasting, and predictive modeling. Builds and evaluates models in Jupyter and Google Colab using pandas, NumPy, scikit-learn, TensorFlow/Keras, SHAP, AIF360, and OpenAI API workflows, emphasizing reproducibility, cross-validation, benchmark comparison, model interpretation, and evidence-based limitations. Brings prior financial-services and compliance operations experience involving sensitive data, documentation accuracy, process discipline, and independent remote work.

TECHNICAL SKILLS

Programming & Data Analysis: Python, pandas, NumPy, data cleaning, exploratory data analysis (EDA), feature preparation, structured data handling, CSV/JSON

Machine Learning & Predictive Modeling: scikit-learn, classification, regression, clustering, Logistic Regression, Random Forest, Gradient Boosting, train/test evaluation, cross-validation, model comparison

Explainable & Responsible AI: SHAP, model explainability, fairness evaluation, AIF360, Reweighting, Adversarial Debiasing, statistical parity difference, disparate impact, equal opportunity difference

NLP & Generative AI: OpenAI API, GPT-based workflows, natural language processing, transformer concepts, prompt-based evaluation, source-fidelity and response-quality analysis

Deep Learning & AI Methods: TensorFlow/Keras, convolutional neural networks (CNNs), reinforcement learning, Q-learning, planning graphs, breadth-first search, constraint satisfaction

Forecasting, Metrics & Tools: ARIMA, seasonal-naive benchmarking, ROC-AUC, F1, RMSE, R-squared (R²), Jupyter Notebook, Google Colab, Matplotlib, SQL fundamentals, relational databases, Microsoft Excel

SELECTED AI & DATA PROJECTS

Explainable AI & Predictive Modeling with SHAP

Python | scikit-learn | SHAP | Random Forest | IT 4538 Advanced AI Coursework

- Built a 100-tree Random Forest on California Housing data and achieved held-out R² of 0.8051, RMSE of 0.5053, and MAE of 0.3275; 5-fold cross-validation R² averaged 0.8045 +/- 0.0068.
- Used SHAP TreeExplainer for global and local model auditing, feature-attribution analysis, and extreme-error review while distinguishing explanation from causal or fairness claims.

Bias Detection & Mitigation in Income Prediction

Python | AIF360 | Logistic Regression | Responsible AI | IT 4538 Advanced AI Coursework

- Evaluated the UCI Adult dataset with a Logistic Regression baseline that achieved 85.38% accuracy and ROC-AUC of 0.9099; measured female and male favorable-outcome rates of 11.37% and 31.38%.
- Compared Reweighting and Adversarial Debiasing using statistical parity difference, disparate impact, equal opportunity difference, accuracy, and F1 to document fairness-performance trade-offs.

Healthcare Patient Outcome Predictive Modeling

Python | scikit-learn | Random Forest | UCI Heart Disease | IT 4538 Advanced AI Coursework

- Developed and evaluated a Random Forest classification pipeline with 88.52% test accuracy, ROC-AUC of 0.9334, 92.86% sensitivity, F1 of 88.14%, and 5-fold cross-validation ROC-AUC of 0.8888.
- Reviewed false positives and false negatives, calibration uncertainty, fairness, and limitations rather than treating strong test metrics as evidence of clinical readiness.

Generative AI & Natural Language Processing Evaluation

Python | OpenAI API | NLP | IT 4538 Advanced AI Coursework

- Designed and executed a controlled GPT-based NLP workflow using the Gettysburg Address as source text, evaluating fluency, source fidelity, instruction compliance, and observed limitations while implementing secure API-key handling and reproducible procedures.

SELECTED AI & DATA PROJECTS - CONTINUED

Time-Series Sales Forecasting & Benchmark Evaluation

Python | ARIMA | Time-Series Analysis | IT 4250 Data Analytics & AI in the Cloud

- Compared ARIMA with a seasonal-naive baseline on monthly retail sales; the seasonal-naive method outperformed ARIMA on the held-out evaluation window, supporting evidence-based model selection and careful interpretation of a short time series.

AI Planning Agent for Room Navigation

Python | Planning Graphs | Breadth-First Search | IT 4535 Introduction to Artificial Intelligence

- Modeled room-to-room state transitions and used breadth-first search to extract a 4-action plan from Room A to Room F; the executed run expanded 6 states, generated 11 actions, and built 9 planning levels.

Additional Project Experience: N-Queens constraint satisfaction (backtracking vs. forward checking); Iris classification and clustering; housing regression comparison; CIFAR-10 CNN image classification; Q-learning grid-world agent; GAN image generation.

EDUCATION

Capella University

Bachelor of Science in Information Technology

Specialization: Data Analytics and Artificial Intelligence

Expected Graduation: **September 2028** | GPA: **4.0**

Relevant Coursework: Advanced Artificial Intelligence; Data Analytics and AI in the Cloud; Introduction to Artificial Intelligence; Python Programming; Relational Databases; Statistics

PROFESSIONAL EXPERIENCE

Working Solutions | Outbound Engagement Specialist

Glendale, AZ | May 2024 - Aug 2024

- Managed structured outbound engagement and recordkeeping in a remote environment while independently following established workflows and performance requirements.
- Maintained accurate activity documentation and consistent quality standards across high-volume, process-driven work.

NexRep LLC | Customer Care Representative

Peoria, AZ | Apr 2020 - Jun 2021

- Managed high-volume member interactions while verifying information, updating records, and following HIPAA-related procedures in a remote environment.
- Navigated multiple systems to resolve service issues while maintaining accurate documentation and protecting sensitive information.

Aerotek | Collections Care Representative III

Phoenix, AZ | Aug 2018 - Mar 2019

- Reviewed account and delinquency information, analyzed outstanding balances, and documented payment arrangements in a metrics-driven financial-services environment.
- Applied procedural and compliance requirements while handling sensitive financial data and maintaining accurate account records.

Carver Federal Savings Bank | Universal Banker

Brooklyn, NY | Feb 2017 - May 2018

- Reviewed and verified customer and account information, investigated discrepancies, and supported account opening, maintenance, and transaction processing.
- Maintained documentation accuracy and regulatory awareness while working with confidential financial information.

Visa Application Centre for the Consulate General of Canada | Application Analyst

New York, NY | Jan 2014 - Jan 2015

- Reviewed applications and supporting documentation for completeness, accuracy, and adherence to established procedural requirements.
- Processed sensitive records in a detail-intensive environment requiring consistent quality control and confidentiality.

Additional Financial Services & Customer Operations Experience

JPMorgan Chase; Virtual Office Partners | 2009 - 2013

- Developed foundational experience in financial transactions, customer and account records, workflow management, confidential data handling, and structured operational procedures.